The exchange criteria of catalyst blocks against the reading of the

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device based on a defined calibration procedure and/or exchange requirements for the device. The justified frequency of monitoring should be stated by the applicant.

The exchange criteria of catalyst blocks against the reading of the NOx measurement device are to be specified by the applicant as well as the maintenance, service and calibration requirements for the NO x measurement device. The criteria should ensure timely exchange of the catalyst blocks.

Depending on the proposed onboard verification procedure for the assessment of catalyst condition/degradation, an allowance may be given according to section 7.5 of these Guidelines. Generated alarms or failure codes, in case of exceeded threshold values a s defined by the applicant, are to be provided;

.2 where a strategy without a NOx measurement device is applied, the applicant should additionally specify periodical spot checks as the method to assess the NOx reduction rate as an indicator for catalyst condition/degradation. The applicant is to provide the details of:

.1 the expected deterioration curve under expected operating conditions or the life of the catalyst under expected

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condition/degradation. The applicant is to provide the details of:

.1 the expected deterioration curve under expected operating conditions or the life of the catalyst under expected operating conditions;

.2 factors which can influence catalyst NO x reduction efficiency; and

.3 guidance on how to assess catalyst NO x reduction efficiency based on periodical spot checks or monitoring as specified by the applicant, if applicable; records are to be kept for inspection during annual, intermediate and renewal surveys. The frequency of periodical spot checks is to be defined by the applicant considering the expected deterioration of the catalyst. The frequency for spot-checks should be after installation and at least once every 12 months.

Periodical spot checks do not need to be witnessed by the Administration. In cases where spot checks are required,

the checks should be performed at least at 50% of the rated power (for propulsion engines, 75% is preferable), and the guidance on how to assess catalyst condition/degradation should include the following items:

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power (for propulsion engines, 75% is preferable), and the guidance on how to assess catalyst condition/degradation should include the following items:

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.1 procedure for spot checks:
.1 details of NOx measurement device including calibration requirements . NOx measurement device should meet the requirements of appendix III of the NTC 2008;
.2 performance of zero and span check;
.3 test condition (e.g. power and speed setting ranges as well as other applicable engine and SCR settings);
.4 a test report template for the data to be recorded;
.5 sampling probe position(s) for NOx measurement;
.6 test procedures including time duration for "engine with SCR" stabilization and the NOx emission measurement; and
.7 records and results of spot checks should be kept and logged in the record book of engine parameters and should be made available during the initial, annual, intermediate and renewal surveys.
.2 criteria to assess catalyst NOx reduction rate in

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be kept and logged in the record book of engine parameters and should be made available during the initial, annual, intermediate and renewal surveys.
.2 criteria to assess catalyst NOx reduction rate in accordance with the specification as provided by the applicant; and
.3 other strategies on monitoring the catalyst condition/degradation are subject to the approval of the Administration. These strategies

should be accepted only if they cover the entire SCR chamber with all catalyst blocks installed. Testing of single catalyst blocks after removing them from the SCR chamber should not be considered as representative for the entire SCR system;
.9 controlling arrangements and settings of the SCR, e.g. model, specification of control device. This is to include, but not be limited to:
.1 the reductant injection control strategy should include whether it is a feed -forward reductant injection control or feedback reductant injection control strategy;
.2 instrumentation and sensors which are part of the SCR control arrangement, as applicable;
.3 crew instructions for allowable adjustment of control parameters

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injection control strategy;
.2 instrumentation and sensors which are part of the SCR control arrangement, as applicable;
.3 crew instructions for allowable adjustment of control parameters including details of how to prevent unauthorized alteration of the system configuration parameters, programmable logic controller (PLC) data, and central processing units (CPU) as applicable;
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 9
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.4 where a NO x measurement device is used, the following details should be included:
.1 type/model (identification number);
.2 calibration, zero and span check procedures and the periodicity of such checks, if applicable;
.3 zero and span gases to be carried on board if applicable; and
.4 servicing, maintenance and/or exchange requirements;
.5 where the engine system fitted with SCR has different operating modes (e.g. modes for Tier II and Tier III compliance separately), details of the control philosophy for selecting different modes of operation and recording the mode of operation together with means

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modes (e.g. modes for Tier II and Tier III compliance separately), details of the control philosophy for selecting different modes of operation and recording the mode of operation together with means of changing between modes; and
.6 auxiliary control devices, as mentioned in regulation 13.9 and defined in regulation 2.4 of MARPOL Annex VI, respectively, may be used on engine systems fitted with SCR, covering starting and stopping, low load operation and reversing operation, subject to the approval of the Administration;
.10 measures to minimize reductant slip. The maximum reductant slip may be specified by the applicant. Supporting information, including reductant

injection rates under certain engine loads, the catalyst temperature or exhaust gas temperature when reductant in jection occurs, etc. may be included in order to prevent reductant slip from exceeding the specified maximum level. Reductant slip monitoring in the exhaust duct downstream of the SCR or an equivalent means may be accepted as a means to minimize reductant slip. Alternatively, means of alleviating reductant slip (for example,

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maximum level. Reductant slip monitoring in the exhaust duct downstream of the SCR or an equivalent means may be accepted as a means to minimize reductant slip. Alternatively, means of alleviating reductant slip (for example, through the use of an ammonia slip catalyst or active catalyst thermal management) may be accepted as a means to minimize reductant slip;
.11 where the parameter check method is used as the verification procedure: For systems without NOx measurement devices, the applicant should provide details of the relationship between engine load and reductant consumption and the means of checking that reductant flow is appropriate. The Technical File should include proposals for maintaining records o f reductant consumption and also reductant composition and quality. Records of reductant composition and quality may be based on delivery notes where these delivery notes include reductant concentration and quality parameters.